Big Data Management — P1 Report

Cymatics: Sound-Driven Pattern Generation & Analysis Pipeline

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1 Instructions

Repository access: All source code and files will be available on GitHub.

2 Contextualization

2.1 Domain and Problem Statement

This project addresses the **Acoustics & Cymatics** domain, specifically focusing on transforming audio signals into visual cymatics patterns — the geometric shapes that emerge when sound vibrates a physical medium such as water or a metal plate (via sand).

> *"Given an audio signal (recorded live or sourced from datasets/APIs), generate a visual representation of its cymatics pattern (image and video showing its evolution), extract acoustic features, and organize all artifacts into a structured, queryable dataset for analysis."*

Cymatics bridges physics, music, and visual art. While the phenomenon has been studied since Ernst Chladni's 18th-century experiments, there is no large-scale, structured dataset that pairs audio recordings with their corresponding cymatics visualizations and acoustic metadata. This project aims to fill that gap by building an automated pipeline that ingests audio, generates cymatics artifacts, and stores everything in a well-organized data lakehouse.

2.2 Business Value

The pipeline delivers value across multiple domains:

- **Scientific Research:** Researchers in acoustics and physics can study frequency–pattern relationships at scale, comparing how different frequencies, amplitudes, and waveforms produce distinct geometric structures.
- **Music & Sound Design:** Audio engineers and musicians can visualize the "shape" of their sounds, enabling a new modality for understanding and comparing audio content.
- **Art & Creative Applications:** Cymatics patterns are visually striking. The pipeline can generate high-resolution artwork from any audio source, with potential applications in generative art, NFTs, and visual installations.
- **Education:** Interactive cymatics visualizations make abstract concepts like wave interference, resonance, and standing waves tangible for students.
- **Pattern Classification:** The generated dataset enables training ML models to classify sounds or visual patterns — essentially "seeing" sound.

2.3 Data Sources and Categories

The pipeline processes structured, semi-structured, and unstructured data.

Table 1: Data Sources and Categories

Category	Data	Format	Source
Structured	Audio metadata columns (frequency, amplitude, RMS, duration, file paths, . . .)	CSV / Parquet	Generated by the pipeline/Expanding source dataset (if any)
Semi-structured	Per-observation metadata (peak frequencies, spectral peaks, ...)	JSON	Generated by Kafka
Semi-structured	Streaming Event Messages when using Kafka in warm/hot paths	JSON	Generated by Kafka
Unstructured	Raw audio recordings	.wav	Live mic recording, FreeSound API, audio datasets, . . .
Unstructured	Cymatics images (peak frequency pattern)	.png	Generated by the pipeline
Unstructured	Cymatics videos (patterns appearing)	.mp4	Generated by the pipeline

Data Sources

1. **Live Microphone Recording** — the user records audio directly, and images and videos are generated and stored along with their metadata. This represents a warm/near-real-time ingestion path.

2. **Live Visualization (Hot Path)** — user is also able to visualize and see the patterns appearing and changing in real time via live microphone recording and store batches of audio observations and corresponding metadata as the microphone records (Hot Path).

3. **FreeSound / other APIs** — programmatic access to a large repository of Creative Commons audio files, enabling batch ingestion of diverse sound samples (musical instruments, environmental sounds, speech, etc.).

4. **Local Datasets** — pre-existing audio collections (e.g., UrbanSound8K, ESC-50, or custom recordings) loaded via batch processing.

One observation in the final dataset consists of:

- Raw audio file (.wav)
- Generated cymatics image (.png)
- Generated cymatics video (.mp4)

- A JSON metadata file containing: dominant frequency (Hz), all detected peak frequencies, amplitude, RMS, sample rate, duration, ...
- Streaming event messages that appeared if Kafka was used (usually JSON).
- Textual columns providing audio/visualization features, file paths (CSV, Parquet).

2.4 Cymatics Generation Engine

We implement the **PennDesign wave-source interference method**:

1. **Wave Sources:** (audio out points) are placed evenly around a circular plate perimeter (16 sources).
2. Each source emits waves at the detected spectral frequencies:
$$\text{displacement} = \text{amp} \times \sin\left(2\pi \times \text{freq} \times \frac{\text{distance}}{\text{size}}\right)$$
Alternating sine and cosine functions are applied between sources.
3. **Interference Pattern:** is computed as the vector sum of all wave displacements at every point on the grid.
4. **Nodal Lines Detection:** points where displacement ≈ 0, these are where sand/particles would accumulate in a real Chladni plate, forming geometric patterns.
5. **Rendering:** Two-color visualization with zero-crossing lines for geometric detail.

3 Architecture Design

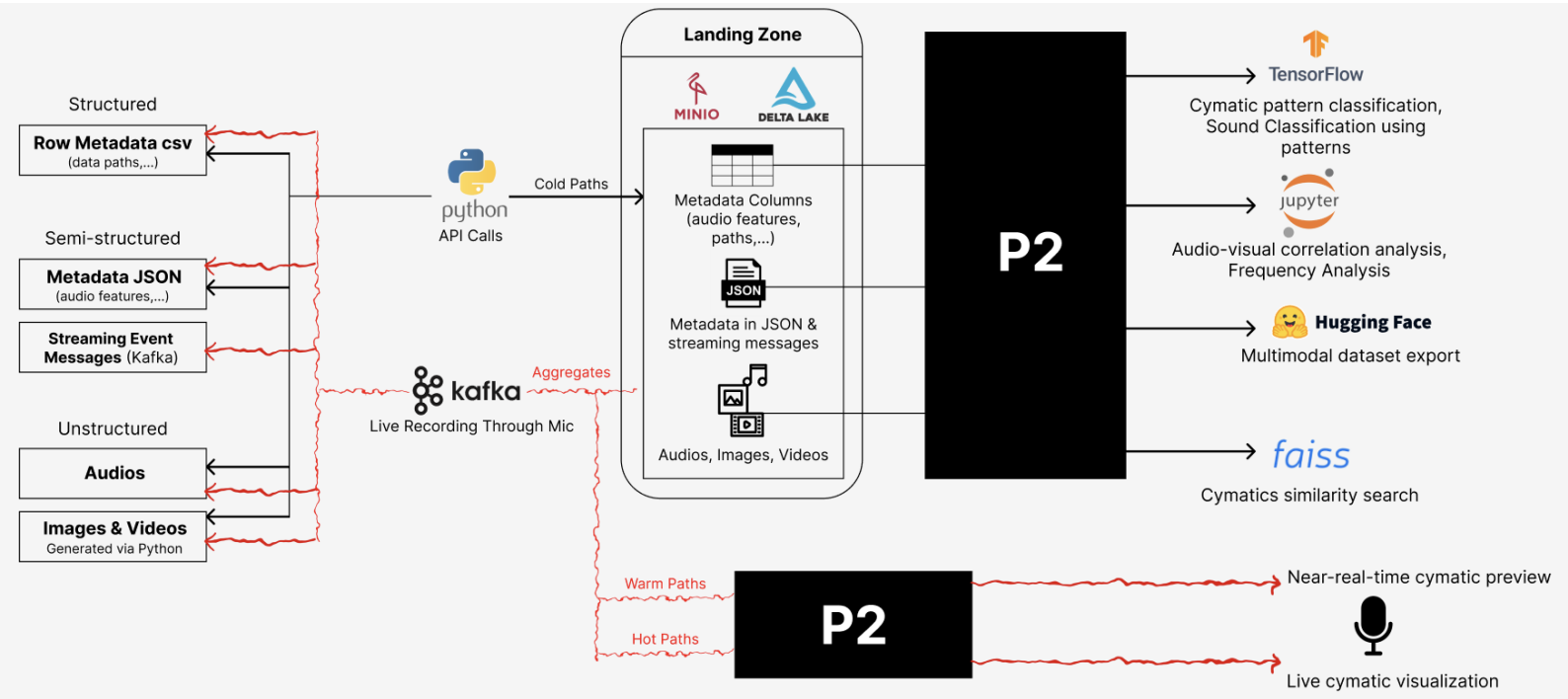

Figure 1: Architecture Design